



DUDEKULA
NAGAJYOTHI
COMET FWC 068

31st International Mathematical Olympiad

Beijing, China

Day I

July 12, 1990

1. Chords AB and CD of a circle intersect at a point E inside the circle. Let M be an interior point of the segment EB . The tangent line at E to the circle through D , E , and M intersects the lines BC and AC at F and G , respectively. If

$$\frac{AM}{AB} = t,$$

find

$$\frac{EG}{EF}$$

in terms of t .

2. Let $n \geq 3$ and consider a set E of $2n - 1$ distinct points on a circle. Suppose that exactly k of these points are to be colored black. Such a coloring is “good” if there is at least one pair of black points such that the interior of one of the arcs between them contains exactly n points from E . Find the smallest value of k so that every such coloring of k points of E is good.

3. Determine all integers $n > 1$ such that

$$\frac{2^n + 1}{n^2}$$

is an integer.

31st International Mathematical Olympiad

Beijing, China

Day II

July 13, 1990

4. Let $\mathbb{Q}^+$ be the set of positive rational numbers. Construct a function

$$f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$$

such that

$$f(xf(y)) = \frac{f(x)}{y}$$

for all x, y in $\mathbb{Q}^+$.

5. Given an initial integer $n_0 > 1$, two players, A and B , choose integers $n_1, n_2, n_3, \dots$ alternately according to the following rules:

Knowing n_{2k} , A chooses any integer n_{2k+1} such that

$$n_{2k} \leq n_{2k+1} \leq n_{2k}^2.$$

Knowing n_{2k+1} , B chooses any integer n_{2k+2} such that

$$\frac{n_{2k+1}}{n_{2k+2}}$$

is a prime raised to a positive integer power.

Player A wins the game by choosing the number 1990; player B wins by choosing the number 1. For which n_0 does:

- (a) A have a winning strategy?
 - (b) B have a winning strategy?
 - (c) Neither player have a winning strategy?
6. Prove that there exists a convex 1990-gon with the following two properties:
- (a) All angles are equal.
 - (b) The lengths of the 1990 sides are the numbers

$$1^2, 2^2, 3^2, \dots, 1990^2$$

in some order.